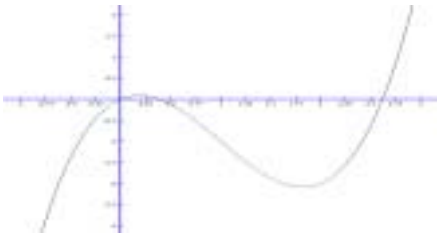
Question	Answer	Marks	Partial Marks
3(a)	0 –2 0.9	**3**	**B1** for each
3(b)	Correct curve 	**4**	**B3 FT** for 9 or 10 points or **B2 FT** for 7 or 8 points or **B1 FT** for 5 or 6 points
3(c)	−0.45 to –0.35 1 2.35 to 2.45	**3**	**FT** *their* graph **B1** for each in the correct position If zero scored, **SC1FT** for 3 correct values
3(d)(i)	$y = 1 - x$ oe	**2**	**B1** for $y = 1 - kx$ oe, $k \neq 0$ or $y = k - x$ oe or $1 - x$
3(d)(ii)	Correct ruled line and 2.25 to 2.4	**3**	**B2FTdep** for correct ruled line or **B1 dep** for line through (0, 1) when extended but not $y = 1$ or with gradient –1.1 to –0.9 or correct line but freehand or **SC2** for $y = x - 1$ ruled after answer [$y =$] $x - 1$ in **(d)(i)** **and** **B1** for 2.25 to 2.4
3(e)	Correct tangent and 1.7 to 3.7	**3**	No daylight between tangent and curve at $x = -0.25$. Point of contact is the midpoint between two vertices of daylight and this point of contact must be between –0.35 and –0.15 **B2** for close attempt at tangent at $x = -0.25$ and answer in range OR **B1** for ruled tangent at $x = -0.25$, no daylight Consider point of contact as midpoint between two vertices of daylight, the midpoint must be between $x = -0.35$ and –0.15 and **M1** dep on **B1** or close attempt at tangent at $x = -0.25$ for $\frac{rise}{run}$
4(a)	100.2 nfww	**4**	**M1** for midpoints soi 65, 80, 95, 105, 112.5, 120 **M1** for use of $\sum fx$ with x in correct interval including both boundaries **M1**dep for $\sum fx \div 180$ dep on previous **M1**
4(b)	0.8 2.8 0.65	**3**	**B1** for each If zero scored, **SC1** for 1.6, 5.6 and 1.3 seen

Question	Answer	Marks	Partial Marks
4(c)	8 34 69 136 164	**2**	**B1** for one error **FT** other values or for 3 or 4 correct
4(d)	Correct diagram	**3**	**B1FT** for correct vertical placement for 6 plots **B1** for correct horizontal placement for 6 plots **B1FT dep on at least B1** for reasonable increasing curve or polygon through *their* 6 points If zero scored, **SC1FT** for 5 out of 6 correct plots
4(e)(i)	15 to 17	**2**	**B1** for [LQ =] 93 to 94 or [UQ =] 109 to 110
4(e)(ii)	107 to 109	**2**	**B1** for 126 seen
4(e)(iii)	66 to 72	**2**	**FT** their graph for 2 marks **B1** for answer 106 to 114 or **B1FT** *their* graph reading at 106 cm seen
5(a)(i)	[h =] $253.8 \div 18 \div \left(\frac{6}{2}\right)$ or [h =] $\frac{253.8 \times 2}{6 \times 18}$ or [h =] $\frac{253.8}{18 \times \frac{6}{2}}$	**3**	For **M3** no errors at any stage **M2** for 253.8 = $\frac{1}{2} \times 6 \times h \times 18$ oe (no previous errors) or **M1** for triangle area = $\frac{1}{2} \times 6 \times h$ soi
5(a)(ii)	38.1 or 38.06 to 38.08	**2**	**M1** for tan = $\frac{4.7}{6}$ oe
5(b)	358 or 357.9 to 358	**6**	**M1** for $6^2 + 4.7^2$ **M1** for $\sqrt{6^2 + 4.7^2} \times 18$ [× 2] **M1** for 6 × 18 [× 2] **M1** for 4.7 × 18 **M1** for $2 \times \frac{1}{2} \times 6 \times 4.7$ oe
6(a)(i)	14	**1**	
6(a)(ii)	16	**1**	
6(a)(iii)	$\frac{20}{462}$ oe	**3**	**M2** for $\frac{5}{22} \times \frac{4}{21}$ or **M1** for $\frac{5}{22}$ seen